



Prayas JEE (2024)

Binomial Theorem Revision Practice Sheet

JEE

Single Correct Type Questions: (1 to 15)

1. The coefficient of second term in the binomial expansion of $\left(x^2 + \frac{3}{x^6}\right)^4$ is

(1) 4 (2) 8
(3) 12 (4) 18

2. The greatest number of which $6^n - 5n - 1$ is always divisible, is (where $n \in N$)

(1) 5 (2) 25
(3) 50 (4) 75

3. Find the value of

$$\frac{16^3 + 9^3 + 3 \times 16 \times 9 \times 25}{3^6 + 6 \times 243 \times 2 + 15 \times 81 \times 4 + 20 \times 27 \times 8 + 15 \times 9 \times 16 + 6 \times 3 \times 32 + 64}$$
 is

(1) 1 (2) 2
(3) 3 (4) None of these

4. The 6th terms in the expansion of $\left(2x - \frac{1}{3x}\right)^{10}$ is

(1) $-\frac{896}{27}x$ (2) $-\frac{896}{27}$
(3) $-\frac{448}{81}x$ (4) $-\frac{448}{81}$

5. Find 'a' if the 17th and 18th terms of the expansion $(2 + a)^{50}$ are equal :-

(1) 0 (2) -1
(3) 1 (4) None of these

6. Find 'n' if the ratio of the fifth form from the beginning to the fifth term from the end in the expansion of $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$ is $\sqrt{6} : 1$

(1) 10 (2) 11
(3) 9 (4) None of these

7. Find the term in $\left(\sqrt{\frac{a}{3b}} + \sqrt[3]{\frac{b}{\sqrt{a}}}\right)^{21}$

Which has the same power of 'a' and 'b'.

(1) 11 (2) 12
(3) 13 (4) 14

8. The number of term in the expansion of $\left(x^{2/3} + \sqrt{3}\right)^{20}$ are

(1) 20 (2) 21
(3) 22 (4) 19

9. The number of terms in the expansion of $(1 + 2x^2 + x^4)^{20}$ are

(1) 40 (2) 80
(3) 41 (4) 81

10. The number of terms in the expansion of $(x + y + z)^{10}$ are

(1) 55 (2) 56
(3) 65 (4) 66

11. The coefficient of x^{13} in the expansion of $(1 - x)^5 \times (1 + x + x^2 + x^3)^4$ is

(1) 2 (2) 5
(3) 4 (4) 8

12. The term which is independent of 'x' in the expansion

of $(1 + x + 2x^3) \left[\frac{3x^2}{2} - \frac{1}{3x} \right]^9$ is

(1) $\frac{17}{54}$ (2) $\frac{7}{18}$
(3) $-\frac{2}{27}$ (4) None of these



13. The value of the sum ${}^{17}C_0 + {}^{17}C_1 + {}^{17}C_2 + \dots + {}^{17}C_8$ is

- (1) 2^{14} (2) 2^{18}
(3) 2^{17} (4) 2^{16}

14. If $(1 + x - 2x^2)^{20} = a_0 + a_1x + a_2x^2 + \dots + a_{40}x^{40}$ then the value of $a_1 + a_3 + a_5 + \dots + a_{39}$ is

- (1) 2^{18} (2) -2^{18}
(3) 2^{19} (4) -2^{19}

15. The value of the sum $\sum_{r=1}^n \frac{r \cdot {}^nC_r}{r-1}$ is

- (1) $\frac{n(n-1)}{2}$ (2) $\frac{n(n+1)}{2}$
(3) $n(n-1)$ (4) None of these

More Than One Correct Option: (16 to 18)

16. If the sum of coefficients in the expansion of $(x - 2y + 3z)^n$ is 128, then the greatest coefficients in the expansion of $(1 + x)^n$ is(are)

- (1) 7C_2 (2) 7C_3
(3) 7C_4 (4) 7C_5

17. If the first three terms in the expansion of $(a + b)^n$ are 729, 7290 and 30375 then

- (1) $a = 3, b = 5$ (2) $a = 4, n = 6$
(3) $b = 5, n = 5$ (4) $b = 5, n = 6$

18. The 10th term of $\left(3 - \sqrt{\frac{17}{4} + 3\sqrt{2}}\right)^{20}$ is

- (1) an irrational number
(2) a rational number
(3) a positive number
(4) a negative number

Passage Type Questions: (19 to 20)

If $a = {}^{20}C_0 + {}^{20}C_3 + {}^{20}C_6 + {}^{20}C_9 + \dots$, $b = {}^{20}C_1 + {}^{20}C_4 + {}^{20}C_7 + \dots$ and $c = {}^{20}C_2 + {}^{20}C_5 + {}^{20}C_8 + \dots$ then

19. Value of $a^3 + b^3 + c^3 - 3abc$ is

- (1) 2^{10} (2) 2^{20}
(3) 0 (4) 2^{19}

20. Value of $(a - b)^2 + (b - c)^2 + (c - a)^2$ is

- (1) 1 (2) 2
(3) 2^{20} (4) 2^{40}

Match the Column Type Questions: (21 to 23)

21. Consider set

$$A = \{T(r+1) = {}^nC_r(3)^{n-r}(5x)^r, r = 0, 1, 2, 3, \dots, n\}$$

	List -I		List-II
(A)	For $x = 1$ if $T(12)$ is the greatest then the value of 'n' can be	(P)	10
(B)	For $n = 25$, if $T(21)$ is the greatest then the value of $5x$ can be	(Q)	18
(C)	For $n = 20$, if $T(10) = T(11)$ then the value of $33x$ can be	(R)	17
(D)	For $x = 2$, if $T(8)$ is the second largest then the value of 'n' is	(S)	12

	A	B	C	D
(1)	P	Q	R	S
(2)	Q	P	R	S
(3)	R	S	Q	P
(4)	R	P	Q	S

22. Match the following:

	List -I		List-II
(A)	${}^nC_0 - {}^nC_2 + {}^nC_4 - {}^nC_6 + \dots$	(P)	$2^{n/2} \cos \frac{n\pi}{4}$
(B)	${}^nC_0 + {}^nC_2 + {}^nC_4 + {}^nC_6 + \dots$	(Q)	$2^{n/2} \sin \frac{n\pi}{4}$
(C)	${}^nC_1 - {}^nC_3 + {}^nC_5 - {}^nC_7 + \dots$	(R)	0
(D)	${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots$	(S)	2^{n-1}

	A	B	C	D
(1)	P	S	Q	R
(2)	Q	S	P	R
(3)	P	R	Q	S
(4)	Q	R	P	S



23. Match the following:

	List -I		List-II
(A)	${}^nC_0 + 2{}^nC_1 + 3{}^nC_2 + \dots + (n+1){}^nC_n$	(P)	$\frac{1}{n}$
(B)	$1 \times 2 \times {}^nC_1 + 2 \times 3 \times {}^nC_2 + \dots + n \times (n+1) \times {}^nC_n$	(Q)	$\frac{2^n - 1}{n+1}$
(C)	$\frac{{}^nC_1}{2} + \frac{{}^nC_3}{4} + \frac{{}^nC_5}{6} + \dots$	(R)	$\frac{(n+2)}{2^{n-1}}$
(D)	$\sum_{r=1}^n (-1)^{r-1} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{r} \right) {}^nC_r$	(S)	$\frac{n(n+3)}{2^{n-2}}$

	A	B	C	D
(1)	P	Q	R	S
(2)	R	S	Q	P
(3)	S	R	Q	P
(4)	R	S	P	Q

Integer Type Questions: (46 to 50)

24. If $P = (10 + 3\sqrt{11})^n$ and $f = p - [p]$ where $[\cdot]$ denotes the greatest integer function, then the value of $p(1-f)$ is equal to _____.
25. The coefficient of x^{12} in the expansion of $(1 - x^2 + x^4)^3 (1 - x)^7$ is _____.



ANSWER KEY

1. (3)
2. (2)
3. (1)
4. (2)
5. (3)
6. (1)
7. (3)
8. (2)
9. (3)
10. (4)
11. (3)
12. (1)
13. (4)

14. (4)
15. (2)
16. (2, 3)
17. (1, 4)
18. (1, 4)
19. (2)
20. (2)
21. (3)
22. (1)
23. (2)
24. 1
25. 99



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